

How Do Developers React to RESTful API Evolution?

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Abstract. With the rapid adoption of REpresentational State Transfer (REST), more software organizations expose their applications as RESTful web APIs and client code developers integrate RESTful APIs into their applications. When web APIs evolve, the client code developers have to update their applications to incorporate the API changes accordingly. However client code developers often encounter challenges during the migration and API providers have little knowledge of how client code developers react to the API changes. In this paper, we investigate the changes among subsequent versions of APIs and classify the identified changes to understand how the RESTful web APIs evolve. We study the on-line discussion from developers to the API changes by analyzing the StackOverflow questions. Through an empirical study, we identify 21 change types and 7 of them are new compared with existing studies. We find that a larger portion of RESTful web API elements are changed between versions compared with Java APIs and WSDL services. Moreover, our results show that *adding new methods* in the new version causes more questions and views from developers. However the *deleted methods* draw more relevant discussions. In general, our results provide valuable insights of RESTful web API evolution and help service providers understand how their consumers react to the API changes in order to improve the practice of evolving the service APIs.

Keywords: REST API, API Evolution, StackOverflow, Social Media.

1 Introduction

Nowadays, on-line users can conduct various tasks such as posting text on Twitter¹ through web applications or services. With the rapid emergence of REpresentational State Transfer (REST) and high demand of engaging on-line experience from end-users, software organizations such as Twitter are willing to open their applications as RESTful web service APIs described in plain HTML pages [7]. Client code developers often integrate the web APIs into their applications or services to accelerate their development or stay away from low level programming tasks [4]. Typically web API providers (*e.g.*, Twitter) evolve their APIs for various reasons, such as adding new functionality [10]. Client code developers have no control of web API evolution and have to evolve their

¹ <https://twitter.com/>

client applications or services to incorporate the changes of new versions of web APIs [4][10]. It causes difficulties to developers to migrate client applications, since the API client code developers have no knowledge of how the web APIs evolve [2]. Moreover, API providers are not aware of how client code developers react to web API evolution. Therefore analyzing and understanding client code developers' on-line discussion related to API changes is essential for service providers to improve API evolution practices. The communication gap between API providers and client developers should be resolved.

Recently, several research studies have explored the impact of API evolution. For example, Espinha *et al.* [4] conduct semi-structured interviews with web API client developers and mine source code changes of client applications. Li *et al.* [10] conduct an empirical study on classifying the changes of web API evolution, without studying the developers' reactions to the changes. Understanding all of the possible types of changes can help developers have a better preparation for migration. Recently, crowd-sourced resources such as StackOverflow is getting popular among developers. When studying the new changes of web APIs or solving a specific software development problem, client code developers start posting questions or development experience on these crowd-sourced resources instead of mailing lists or project-specific forums [9][12]. Therefore, crowd-sourced social media platforms become an excellent source for studying developers' discussion. Linares-Vásquez *et al.* [11] investigate how developers react to the Android API instability in StackOverflow², a question & answer website for developers to share experience on software development. They study which types of changes trigger more questions and more discussion in StackOverflow.

In this paper, we conduct an empirical study on API changes between subsequent versions of web APIs to explore all of the possible types of changes during API evolution. Since web API source code is usually not publicly available, we compare web API documentation (*i.e.*, migration guides or reference documents), and identify changes between subsequent versions of each API. We further categorize the changes into different change types. Compared with the change types identified in [10], we identify 7 more types of changes in web API evolution. Moreover, we explore how client code developers react to identified types of changes by analyzing developers' online discussion regarding API changes. We adopt the analysis approach in [11] for analyzing StackOverflow posts related to RESTful APIs. To the best of our knowledge, we are the first to link the analysis of developer discussion in StackOverflow with web API changes.

We conduct our empirical study on 11 web APIs from 9 application domains and address the following three research questions:

RQ1. What are the change types of web API evolution?

Identifying and understanding all of the possible types of changes of RESTful web APIs is useful for client code developers to have a better preparation for migrating their code. We manually compare the API documentation of a web API to identify the changes [10][3]. We classify the identified changes into 21 change types and count the number of changes for each change type. Compared with existing research work, we identify new unreported change types. Our empirical results

² <http://stackoverflow.com/>